

Week - 06 - Graphics\Lecture_11_code_4.py

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1  '''
2  This simulation uses the coordinate system provided by the graphics library.
3  We now divide the screen based on our own unit definition. For example,
4  the following window is 800 x 800 pixels, when we set its coordinates
5  as 0,0,200,200, we get 4-pixels per unit.
6  Also, we can now set x=0, y=0 to be the bottom-left corner which represents
7  a real-world like situation.
8  '''
9
10 from graphics import *
11 import math
12
13 from Animate import animate
14 from Projectile import *
15
16 class Canon:
17     # Now a composition design
18     def __init__(self, cid, length, type):
19         self.cid = cid
20         self.length = length
21         self.type = type
22         self.bomb = Projectile(201, 25, 12)
23         self.win = GraphWin("Projectile Simulation", 800, 800)
24         self.win.setCoords(0, 0, 200, 200)
25
26 # def method_name (self, another_object)
27     def fire(self, velocity, theta):
28         x_vel = velocity * math.cos(theta)
29         y_vel = velocity * math.sin(theta)
30         dt = 0.02
31
32         bomb = Circle(Point(10, 10), 1)
33         bomb.setFill('red')
34         bomb.draw(self.win)
35
36         # Now we do not need a SCALE because we have already scaled our window with
37         setCoords
38         # SCALE = 100
39         while self.bomb.y_pos >= 0:
40             dx = x_vel*dt #* SCALE
41             dy = y_vel*dt #* SCALE
42
43             self.bomb.x_pos += x_vel * dt
44             self.bomb.y_pos += y_vel * dt
45             x_vel -= AIR_DRAG * dt
46             y_vel -= G * dt
47
48             print(dx, dy)
49             bomb.move(dx, dy)
50
51             time.sleep(0.02)
52

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52         self.win.getKey()
53
54     def load_projectile(self):
55         self.bomb.x_pos = 0
56
57 canon = Canon('C101', 45, 'Vehicle-Mounted')
58
59 # the velocity is now more realistic
60 canon.fire(40, math.radians(60))
```